EXPERIMENT:10:

AIM:

To write a C program to encrypt a message using the given Playfair cipher matrix.

CODE:

```
#include <stdio.h>

#include <ctype.h>

#include <string.h>


int main() {

   char m[200], a[5][5] = {

      "MFHIK", "UNOPQ", "ZVWXY", "ELARG", "DSTBC"

   };

   int i, j, r1, c1, r2, c2, x, y;


   printf("Enter message: ");

   fgets(m, sizeof(m), stdin);


   for(i = 0; m[i]; i++)

      if(isalpha(m[i])) m[i] = toupper(m[i]);


   printf("Ciphertext: ");

   for(i = 0; m[i]; ) {

      if(!isalpha(m[i])) { i++; continue; }

      char p = m[i++], q;

      while(m[i] && !isalpha(m[i])) i++;

      if(!m[i]) break;

      q = m[i++];
```

```
        if(p == 'J') p = 'I';
        if(q == 'J') q = 'I';

        for(x=0;x<5;x++) for(y=0;y<5;y++) {
            if(a[x][y]==p) r1=x,c1=y;
            if(a[x][y]==q) r2=x,c2=y;
        }

        if(r1==r2)
            printf("%c%c",a[r1][(c1+1)%5],a[r2][(c2+1)%5]);
        else if(c1==c2)
            printf("%c%c",a[(r1+1)%5][c1],a[(r2+1)%5][c2]);
        else
            printf("%c%c",a[r1][c2],a[r2][c1]);
    }
    return 0;
}
```

OUTPUT:

